Quiz App with Score Tracking

Web Page Project • HTML + CSS + JavaScript

Complete project report with source code

1. Project Overview

This project is a browser-based quiz application built with plain HTML, CSS and JavaScript. It shows one question at a time, checks the answer instantly, tracks the score, and saves the high score and the last five attempts in the browser using localStorage. No backend or installation is needed: the whole app is one file, **index.html**.

Objectives

- Learn DOM manipulation and event handling in JavaScript.
- Learn to store data in the browser with localStorage.
- Build a complete, usable app that can be extended with more features.

Technology Used

Part	Technology	Purpose
Structure	HTML5	Start, quiz and result screens
Styling	CSS3	Card layout, responsive design, dark mode, progress bar
Logic	JavaScript (ES6)	Questions, timer, scoring, screens
Storage	localStorage	High score and attempt history

2. Features

Feature	Description
Categories	General Knowledge, Science, Programming, Tamil Cinema & Culture
Levels	Easy, Medium, Hard or All
Random questions	5 shuffled questions per quiz
Timer	15 seconds per question with a progress bar; auto-marks wrong on timeout
Instant feedback	Correct answer turns green, wrong choice turns red
Scoring	10 points per correct answer + bonus equal to seconds left
High score	Saved permanently in the browser
History	Last 5 attempts with date, category, score
Result summary	Score, correct count and wrong count; Play Again button

3. How the App Works

Application flow

- **Start screen**: user picks category and level. High score and history are loaded from localStorage.
- **Quiz screen**: questions are filtered, shuffled and shown one by one. A timer starts for each question.
- **Answer check**: the clicked option is compared with the stored answer index; the score is updated.
- **Result screen**: final stats are shown, and the high score and history are saved.

Main functions

Function	Role
loadStart()	Fills the category list, shows high score and history
startQuiz()	Filters, shuffles and picks questions; resets state
showQuestion()	Renders the question and option buttons
startTimer()	Runs the 15-second countdown
selectAnswer(i)	Locks options, highlights answers, updates score
next()	Moves to the next question or finishes
finish()	Saves high score and history, shows result

Question format

```
{ cat:"Science", level:"easy",
  q:"Which planet is called the Red Planet?",
  options:["Venus","Mars","Jupiter","Saturn"],
  answer:1 }   // answer = index of correct option (0-3)
```

To add questions, copy this object into the QUESTIONS array in the code.

4. How to Run

- Save the code as **index.html**.
- Double-click the file to open it in Chrome, Edge or Firefox.
- Optional: host it free on GitHub Pages or Netlify to share the link.

5. Future Improvements

- Add more questions or load them from a JSON file / API.
- Login and online leaderboard using Node.js or Flask with a database.
- Sound effects and animations.
- Tamil language interface.
- Review screen showing the correct answers after the quiz.

6. Conclusion

The project covers the core ideas of front-end development in a small, practical app. It runs without any server, remembers the user's scores, and is easy to extend, which makes it a good beginner portfolio project.

7. Complete Source Code (index.html)

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>Quiz App with Score Tracking</title>
<style>
  :root { --bg:#f4f6fb; --card:#fff; --text:#1f2430; --muted:#6b7280; --primary:#4f46e5;
    --ok:#16a34a; --bad:#dc2626; }
  @media (prefers-color-scheme: dark) {
    :root { --bg:#12141c; --card:#1c202b; --text:#eef0f6; --muted:#9aa3b5; --primary:#818cf8; }
  }
  * { box-sizing: border-box; }
  body { margin:0; font-family: system-ui, Arial, sans-serif; background:var(--bg);
    color:var(--text);
         min-height:100vh; display:flex; align-items:center; justify-content:center; padding:16px;
     }
  .card { background:var(--card); width:100%; max-width:560px; border-radius:16px; padding:24px;
          box-shadow:0 8px 30px rgba(0,0,0,.12); }
  h1 { margin:0 0 4px; font-size:1.6rem; }
  .muted { color:var(--muted); font-size:.9rem; }
  .hidden { display:none; }
  select, button { font:inherit; }
  select { width:100%; padding:10px; border-radius:10px; border:1px solid var(--muted);
    background:var(--card); color:var(--text); margin:8px 0 14px; }
  button.primary { background:var(--primary); color:#fff; border:0; padding:12px 20px; border-
    radius:10px; cursor:pointer; width:100%; font-weight:600; }
  .top { display:flex; justify-content:space-between; margin-bottom:10px; font-weight:600; }
  .bar { height:8px; background:rgba(128,128,128,.25); border-radius:99px; overflow:hidden;
    margin-bottom:16px; }
  .bar > div { height:100%; background:var(--primary); width:100%; transition:width 1s linear; }
  .opt { display:block; width:100%; text-align:left; margin:8px 0; padding:12px 14px; border-
    radius:10px;
         border:2px solid rgba(128,128,128,.35); background:transparent; color:var(--text);
    cursor:pointer; }
  .opt:hover:not(:disabled) { border-color:var(--primary); }
  .opt.correct { border-color:var(--ok); background:rgba(22,163,74,.15); }
  .opt.wrong { border-color:var(--bad); background:rgba(220,38,38,.15); }
  .stats { display:grid; grid-template-columns:repeat(3,1fr); gap:10px; margin:16px 0; text-
    align:center; }
  .stats div { background:rgba(128,128,128,.12); border-radius:10px; padding:10px; }
  .stats b { display:block; font-size:1.4rem; }
  ul { padding-left:18px; } li { margin:4px 0; }
</style>
</head>
<body>
<div class="card">

  <!-- START SCREEN -->
  <section id="start">
    <h1>Quiz App</h1>
    <p class="muted">Test your knowledge. Your scores are saved in this browser.</p>
    <label>Category</label>
    <select id="category"></select>
    <label>Level</label>
    <select id="level">
      <option value="all">All levels</option>
      <option value="easy">Easy</option>
```

```
      <option value="medium">Medium</option>
      <option value="hard">Hard</option>
    </select>
    <button class="primary" id="startBtn">Start Quiz</button>
    <p>High score: <b id="highScore">0</b></p>
    <h3>Last 5 attempts</h3>
    <ul id="history"></ul>
  </section>

  <!-- QUIZ SCREEN -->
  <section id="quiz" class="hidden">
    <div class="top"><span id="qCount"></span><span id="timer"></span><span
    id="score"></span></div>
    <div class="bar"><div id="bar"></div></div>
    <h2 id="question"></h2>
    <div id="options"></div>
    <button class="primary hidden" id="nextBtn">Next</button>
  </section>

  <!-- RESULT SCREEN -->
  <section id="result" class="hidden">
    <h1>Quiz Finished!</h1>
    <p id="resultMsg" class="muted"></p>
    <div class="stats">
      <div><b id="rScore">0</b>Score</div>
      <div><b id="rCorrect">0</b>Correct</div>
      <div><b id="rWrong">0</b>Wrong</div>
    </div>
    <button class="primary" id="againBtn">Play Again</button>
  </section>

</div>

<script>
/* ---------- 1. QUESTION BANK ---------- */
const QUESTIONS = [
  { cat:"General Knowledge", level:"easy",   q:"What is the capital of India?",
    options:["Mumbai","New Delhi","Chennai","Kolkata"], answer:1 },
  { cat:"General Knowledge", level:"easy",   q:"How many days are in a leap year?",
    options:["365","364","366","367"], answer:2 },
  { cat:"General Knowledge", level:"medium", q:"Which river is known as the 'Dakshina Ganga'?",
    options:["Kaveri","Vaigai","Krishna","Godavari"], answer:0 },
  { cat:"General Knowledge", level:"hard",   q:"In which year did India become a republic?",
    options:["1947","1948","1950","1952"], answer:2 },
  { cat:"Science", level:"easy",   q:"Which planet is called the Red Planet?",
    options:["Venus","Mars","Jupiter","Saturn"], answer:1 },
  { cat:"Science", level:"easy",   q:"What gas do plants absorb from the air?",
    options:["Oxygen","Nitrogen","Carbon dioxide","Hydrogen"], answer:2 },
  { cat:"Science", level:"medium", q:"What is the chemical symbol for Sodium?",
    options:["So","Na","Sd","N"], answer:1 },
  { cat:"Science", level:"hard",   q:"What is the SI unit of electrical resistance?",
    options:["Volt","Ampere","Ohm","Watt"], answer:2 },
  { cat:"Programming", level:"easy",   q:"What does HTML stand for?", options:["Hyper Text Markup
    Language","High Text Machine Language","Hyperlinks Text Mark Language","Home Tool Markup
    Language"], answer:0 },
  { cat:"Programming", level:"easy",   q:"Which language adds interactivity to web pages?",
    options:["CSS","HTML","JavaScript","SQL"], answer:2 },
  { cat:"Programming", level:"medium", q:"Which CSS property changes text color?", options:["font-
```

```
    color","color","text-color","fg"], answer:1 },
  { cat:"Programming", level:"hard",   q:"What does JSON.parse() do?", options:["Converts object
    to string","Converts string to object","Deletes JSON","Sends JSON to server"], answer:1 },
  { cat:"Tamil Cinema & Culture", level:"easy",   q:"Which festival is the Tamil harvest
    festival?", options:["Diwali","Pongal","Holi","Onam"], answer:1 },
  { cat:"Tamil Cinema & Culture", level:"medium", q:"Who wrote the Thirukkural?",
    options:["Kambar","Ilango Adigal","Thiruvalluvar","Bharathiyar"], answer:2 },
  { cat:"Tamil Cinema & Culture", level:"medium", q:"Which is the classical dance form of Tamil
    Nadu?", options:["Kathak","Bharatanatyam","Odissi","Kuchipudi"], answer:1 },
];

const TIME_PER_QUESTION = 15;  // seconds
const QUESTIONS_PER_QUIZ = 5;

/* ---------- 2. HELPERS ---------- */
const $ = id => document.getElementById(id);
const show = id => { ["start","quiz","result"].forEach(s => $(s).classList.add("hidden"));
    $(id).classList.remove("hidden"); };
const shuffle = arr => [...arr].sort(() => Math.random() - 0.5);

/* localStorage wrapped in try/catch so the app never breaks */
const store = {
  get(key, fallback) { try { return JSON.parse(localStorage.getItem(key)) ?? fallback; } catch {
    return fallback; } },
  set(key, val)      { try { localStorage.setItem(key, JSON.stringify(val)); } catch {} }
};

/* ---------- 3. STATE ---------- */
let list = [], index = 0, score = 0, correct = 0, wrong = 0, timeLeft = 0, tick = null, answered =
     false;

/* ---------- 4. START SCREEN ---------- */
function loadStart() {
  const cats = [...new Set(QUESTIONS.map(q => q.cat))];
  $("category").innerHTML = `<option value="all">All categories</option>` + cats.map(c =>
    `<option>${c}</option>`).join("");
  $("highScore").textContent = store.get("quiz_high", 0);
  const hist = store.get("quiz_history", []);
  $("history").innerHTML = hist.length
    ? hist.map(h => `<li>${h.date} &mdash; ${h.category}: <b>${h.score}</b>
    (${h.correct}/${h.total} correct)</li>`).join("")
    : "<li class='muted'>No attempts yet</li>";
  show("start");
}

function startQuiz() {
  const cat = $("category").value, lvl = $("level").value;
  let pool = QUESTIONS.filter(q => (cat === "all" || q.cat === cat) && (lvl === "all" || q.level
    === lvl));
  if (!pool.length) { alert("No questions for this combination. Try another."); return; }
  list = shuffle(pool).slice(0, QUESTIONS_PER_QUIZ);
  index = score = correct = wrong = 0;
  show("quiz");
  showQuestion();
}

/* ---------- 5. QUIZ SCREEN ---------- */
function showQuestion() {
```

```
  answered = false;
  const q = list[index];
  $("qCount").textContent = `Q ${index + 1}/${list.length}`;
  $("score").textContent = `Score: ${score}`;
  $("question").textContent = q.q;
  $("nextBtn").classList.add("hidden");
  $("options").innerHTML = "";
  q.options.forEach((text, i) => {
    const b = document.createElement("button");
    b.className = "opt"; b.textContent = text;
    b.onclick = () => selectAnswer(i);
    $("options").appendChild(b);
  });
  startTimer();
}

function startTimer() {
  clearInterval(tick);
  timeLeft = TIME_PER_QUESTION;
  $("bar").style.transition = "none"; $("bar").style.width = "100%";
  setTimeout(() => { $("bar").style.transition = "width 1s linear"; }, 20);
  $("timer").textContent = `${timeLeft}s`;
  tick = setInterval(() => {
    timeLeft--;
    $("timer").textContent = `${timeLeft}s`;
    $("bar").style.width = (timeLeft / TIME_PER_QUESTION * 100) + "%";
    if (timeLeft <= 0) selectAnswer(-1);   // time up
  }, 1000);
}

function selectAnswer(i) {
  if (answered) return;
  answered = true;
  clearInterval(tick);
  const q = list[index], btns = [...document.querySelectorAll(".opt")];
  btns.forEach((b, n) => {
    b.disabled = true;
    if (n === q.answer) b.classList.add("correct");
    else if (n === i) b.classList.add("wrong");
  });
  if (i === q.answer) {
    correct++;
    score += 10 + timeLeft;            // bonus points for speed
  } else wrong++;
  $("score").textContent = `Score: ${score}`;
  $("nextBtn").textContent = index === list.length - 1 ? "See Result" : "Next";
  $("nextBtn").classList.remove("hidden");
}

function next() {
  index++;
  index < list.length ? showQuestion() : finish();
}

/* ---------- 6. RESULT + SAVE SCORES ---------- */
function finish() {
  const high = store.get("quiz_high", 0);
  const isNew = score > high;
```

```
  if (isNew) store.set("quiz_high", score);

  const hist = store.get("quiz_history", []);
  hist.unshift({
    date: new Date().toLocaleDateString(),
    category: $("category").value === "all" ? "All" : $("category").value,
    score, correct, total: list.length
  });
  store.set("quiz_history", hist.slice(0, 5));   // keep last 5 only

  $("rScore").textContent = score;
  $("rCorrect").textContent = correct;
  $("rWrong").textContent = wrong;
  $("resultMsg").textContent = isNew ? "New high score! " : `High score: ${Math.max(high,
    score)}`;
  show("result");
}

/* ---------- 7. EVENTS ---------- */
$("startBtn").onclick = startQuiz;
$("nextBtn").onclick = next;
$("againBtn").onclick = loadStart;
loadStart();
</script>
</body>
</html>
```